

What we're doing

Building reusable code (issue #41) that answers, for any candidate protein structure: how structurally close is it to anything in MarinerFold's training set? Candidates are FoldBench monomers, de novo designs, and low-MSA natural proteins we may want in an eval set.

Why

The training set (afdb-24M) is already Foldseek-clustered: every structure has a `struct_cluster_id` and a train/val/test split hashed from it. What we lacked was code to take an EXTERNAL candidate and map it back onto those clusters. The mechanism: foldseek easy-search the candidate against a DB of the training-set cluster representatives (TM-score), then read off the nearest train representative, its TM, and its split.

How it works

query_similarity.py: candidate cifs -> foldseek easy-search (alignment-type 1 = TM-align)
-> join each hit's representative to its split -> per-candidate CSV with the nearest
training rep, its TM-score, the free sequence-identity signal, and a verdict (redundant $\geq$
0.9, same_fold $\geq$ 0.5, else novel_fold). build_db_modal.py builds the representative DB +
split manifest on Modal (all 12,005 afdB-24M shards); fetch_db.py pulls it local.

Full DB built, real measurement

The full representative DB is built (build_db_modal.py on Modal, all 12,005 afdb-24M shards): 1,331,330 cluster representatives, a 2.6 GB foldseek DB, splits 0.980/0.0099/0.0099 (matches the ~0.98/0.01/0.01 split-hash). Querying the FoldBench-100 monomers against the full 1.33M-rep training set: 48 redundant (qtm $\geq$ 0.9), 51 same_fold, 1 novel_fold. So 99/100 FoldBench monomers fall in a fold Marifold trained on; the lone novel is 7xcd_A (qtm 0.485). Median nearest-train qtmscore is 0.895. Unit + real-foldseek integration tests pass.

Results: near-total structural overlap

99/100 FoldBench monomers have a same-fold-or-closer match in training, and 48 are structurally near-identical ($qtm \geq 0.9$). The prototype's 229-rep slice had reported 68 novel / 32 same -- that "novelty" was a sparse-DB artifact, which is why the full build mattered. Crucially the overlap is structural, not sequence: 65 of the 99 matches sit below 0.30 sequence identity to their nearest rep. A sequence-only leakage filter would clear most of these as "novel" while structurally they are near-duplicates of trained folds.

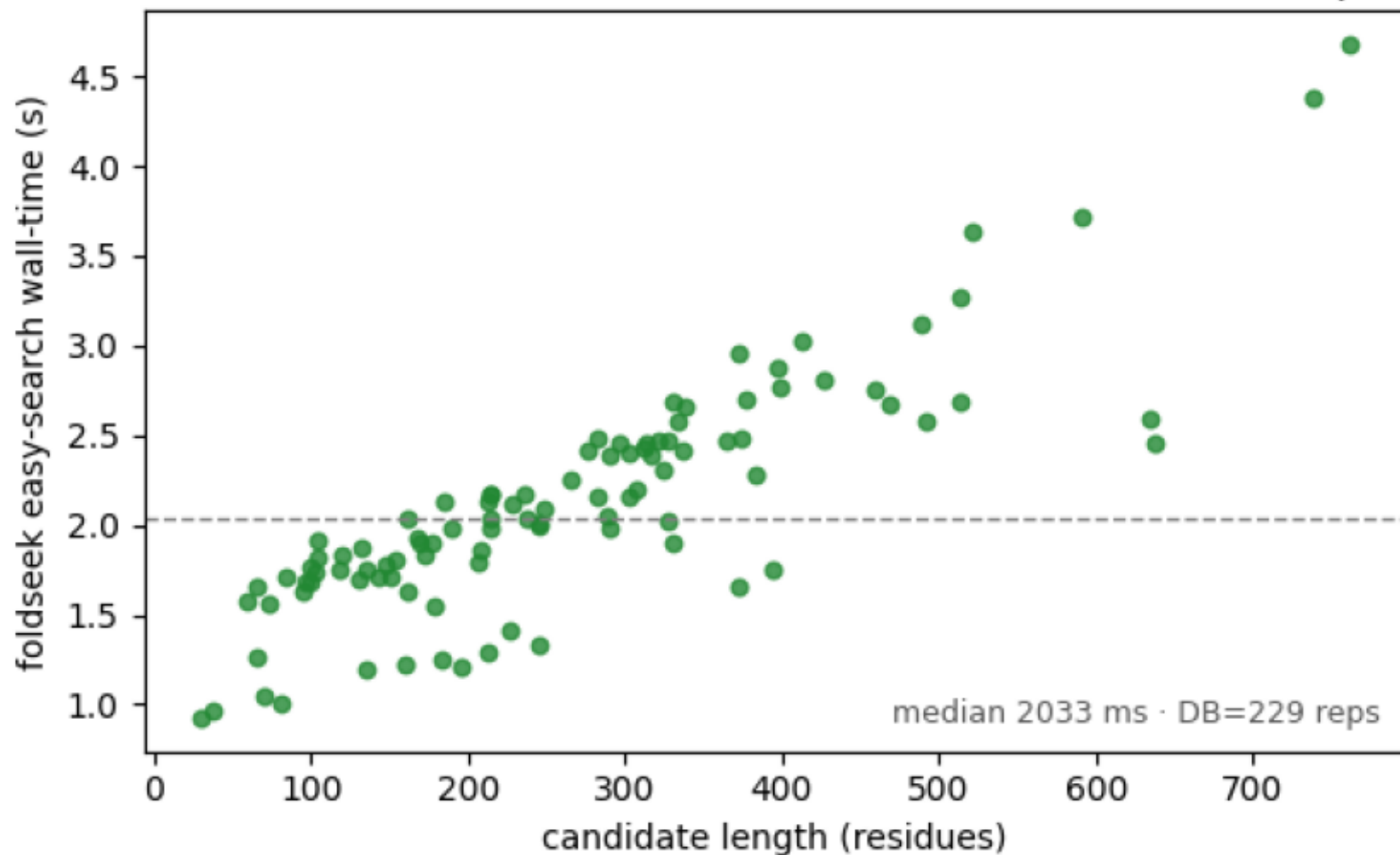
Next

Optional: upload the compact DB + manifest to the HF bucket so others fetch it without rebuilding. Then reuse the same tool to score de novo designs and low-MSA natural proteins, and sub-stratify any eval-set candidate pool by its nearest-train qtmscore (median here 0.895) per `.dev/eval-strategy-summary.md`.

search_timing.png

Per-candidate TM-align search time vs length, against a 229-rep target DB. At this tiny DB size the cost is dominated by Foldseek per-call setup, not DB scaling — absolute numbers will rise with the full ~1.3M-rep training DB. Batched throughput is in the '___ALL___' row.

Per-candidate search time (FoldBench-100 vs afdb-24M full train reps (1.33M))

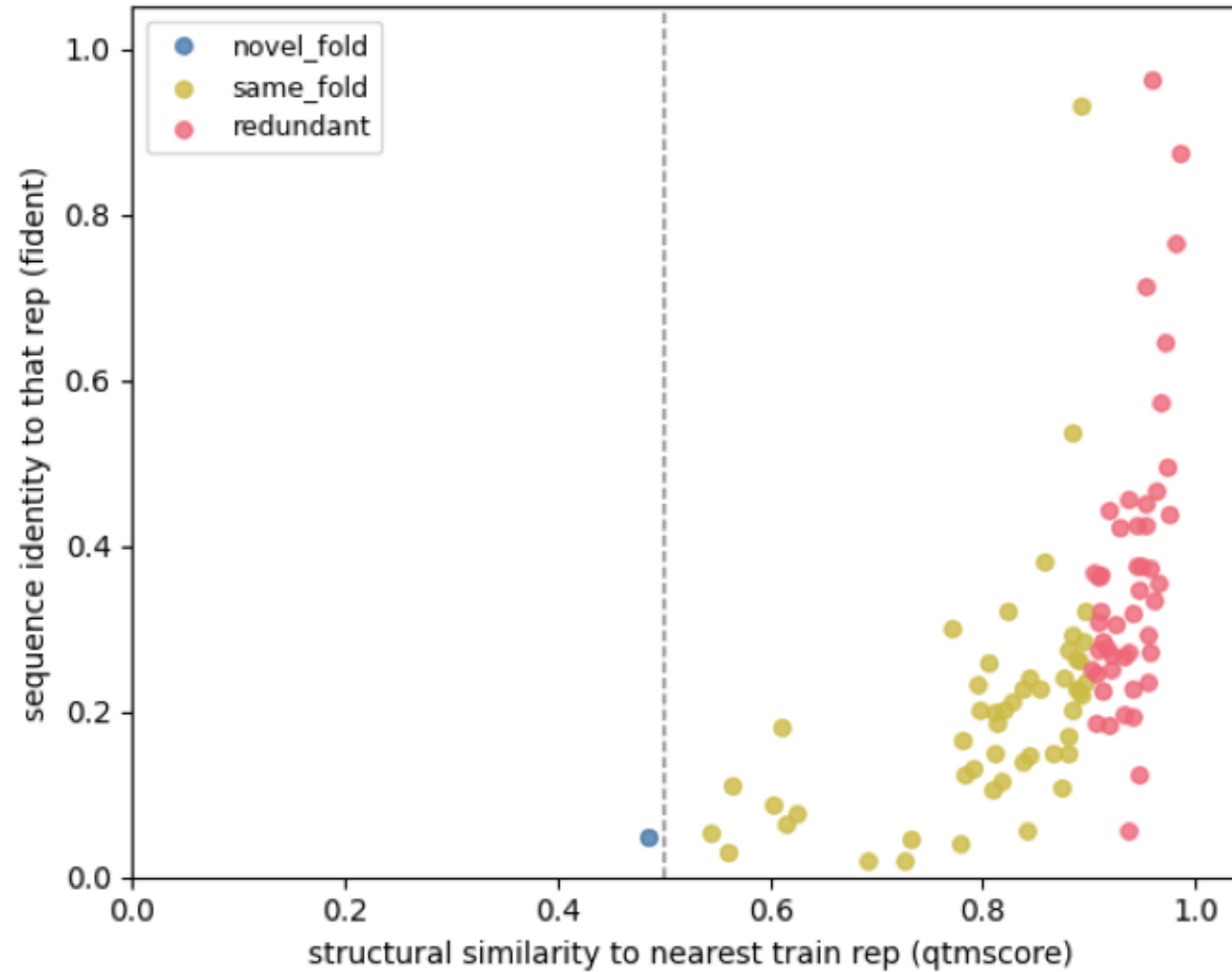


```
$ plot_similarity.py --csv data/foldbench_vs_full_reps_similarity.csv --timings-csv data/timings.csv --suffix FoldBench-100 vs afdb-24M full train reps (1.33M)
```

struct_vs_seq.png

Each candidate's nearest TRAIN representative: structural TM (x) vs sequence identity (y). Bottom-right = structurally close but sequence-distant, the leakage a sequence-only filter would miss. Dashed line = 0.5 same-fold boundary.

Structure vs sequence similarity (FoldBench-100 vs afdb-24M full train reps (1.33M))

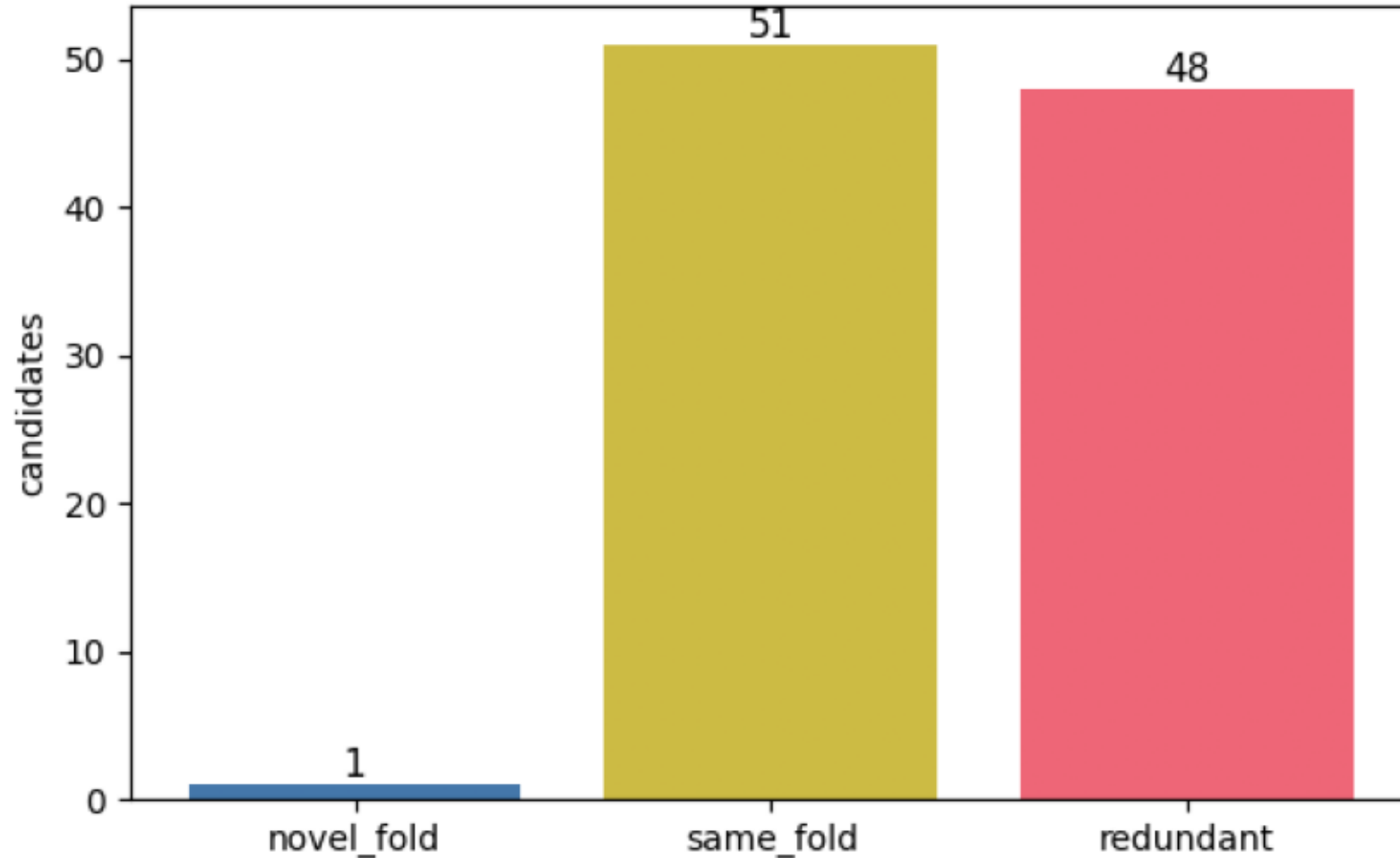


```
$ plot_similarity.py --csv data/foldbench_vs_full_reps_similarity.csv --timings-csv data/timings.csv --suffix FoldBench-100 vs afdb-24M full train reps (1.33M)
```

verdict_hist.png

Per-candidate verdict counts vs the training-set representative DB (FoldBench-100 vs afdb-24M full train reps (1.33M)). Verdict keys off the nearest TRAIN cluster's query-normalized TM-score: redundant ≥ 0.9 , same_fold ≥ 0.5 , else novel_fold.

Train-similarity verdicts (FoldBench-100 vs afdb-24M full train reps (1.33M))



```
$ plot_similarity.py --csv data/foldbench_vs_full_reps_similarity.csv --timings-csv data/timings.csv --suffix FoldBench-100 vs afdb-24M full train reps (1.33M)
```